Jeremy R. Manning
Dartmouth College
Department of Psychological & Brain Sciences
HB 6207 Moore Hall
Hanover, NH 03755

June 12, 2024

To the editors of *PNAS*:

We have enclosed our newly revised manuscript entitled *High-level cognition is supported by information-rich but compressible brain activity patterns* (manuscript tracking number: 2024-00082). We appreciate the reviewers' insightful comments on our previous revision. We provide detailed point-by-point responses to each of the reviewers' comments on the following pages. The reviewers' comments are *italicized* and our responses are in **bold**.

To summarize, Reviewers 1 and 3 both felt the manuscript was suitable for publication in its current form. Reviewer 2 raised several concerns. One set of concerns was related to our interpretations and characterizations of informativeness and compressibility. We have added additional clarifications and explanations to address these concerns. The reviewer also asked for some clarification about several of our methods, and we have added the requested information. Finally, the reviewer suggested several stylistic changes to the abstract and some figures. We have addressed each of these by either implementing the requested changes or by adding relevant explanations to the text or our response letter as appropriate.

Thank you for considering our revised manuscript.

Sincerely,

Jeremy R. Manning
Jeremy.R.Manning@Dartmouth.edu

Reviewer Comments:

Reviewer #1:
Suitable Quality?:Yes
Sufficient General Interest?:Yes
Conclusions Justified?:Yes
Clearly Written?:Yes
Procedures Described?:Yes
Supplemental Material Warranted?:Yes

Comments:
The authors have done an excellent job addressing my concerns. I have no further comments.

We thank the reviewer for their positive assessment of our work.

Reviewer #2:
Suitable Quality?:No
Sufficient General Interest?:Yes
Conclusions Justified?:Yes
Clearly Written?:Yes
Procedures Described?:Yes
Supplemental Material Warranted?:Yes

Comments:
I thank the authors for their response to my prior round of comments. I appreciate, in particular, the addition of the simulations on the relationship between compressibility and informativeness. These add additional context and interpretation to the findings. However, I still have some remaining questions on this work.

- The simulations on the relationship between compressibility and informativeness were interesting. But, while it may be true that they CAN be dissociated, to what extent do the simulations represent scenarios that happen in any real (fMRI) data that is at least somewhat time-locked? Given the correlation between informativeness and compressibility in this dataset, to what extent have the authors shown something novel relative to what has been shown in past papers?

These are interesting questions. There are a few points to unpack. One set of issues concerns time-locking, and how it relates to our measures of informativeness and compressibility. A second set of issues concerns how our simulations might inform our interpretations of "real" data. And a third set of issues concerns how our work relates to other related work. We address each of these sets of issues in turn.

First, in our simulations, the similarities in responses across "participants" are the same in all four synthetic datasets displayed in Fig. 1. To the extent that across-participant similarity reflects stimulus-driven responses, all four of our synthetic datasets capture identical "time-locking" to the "stimulus." But across those different simulated scenarios we can construct data with different levels of "informativeness." So time-locking (e.g., to the stimulus) and informativeness can be decoupled.

The relation between time-locking and compressibility is more subtle. Compressibility depends on the spatial smoothness of the data *as related to stimulus decoding*. Compressibility does not solely reflect spatial autocorrelations (as in the simulations), but rather it reflects the extent to which features can be combined across space *while still retaining sufficient information to decode stimulus timing across people*. In our main analyses, compressibility is intended to get at the notion of different brain areas carrying similar information about when in the story people are listening to. Responses that are independent of the stimulus, or that are entirely person-specific, would not be "compressible" in the way we define it in our study, since across-participant decoding accuracy would be poor regardless of how many features we used to represent the data. This is essentially what we see in the "rest" condition in our study (e.g., purple curves, Fig. 2).

In summary (of these first points), informativeness is intended to capture how well we can decode (across participants) using brain responses, and compressibility is intended to capture redundancies (across brain areas) of the responses. Both of these are indirectly related to time-locking (since without stimulus-related brain activity we would not expect to see informative *or* compressible brain responses), but time-locking alone does not directly imply high or low informativeness, or high or low compressibility. (Illustrating this point was our main reason for running the simulations in Fig. 1.)

Second, the question of whether "real" data behave like synthetic data is difficult to answer. Certainly we conceded that we are not running a complete simulation of the entire brain, or of stimulus-driven responses, in generating those synthetic datasets. Our point in running those simulations in Figure 1 is that informativeness and compressibility refer to different aspects of the data. This is true for synthetic data *and* "real" data.

One could imagine that some real data might display time-locking in ways that are not necessarily captured by our simulations. For example, in the extreme, suppose that brain responses reflected a direct representation of the stimulus itself, plus some noise. In that case, the responses might appear to be "time-locked" across people, as long as everyone experienced the same stimulus and represented it (internally) in a similar way. Whether or not the data showed up as being "informative" would depend on the signal-to-noise ratio in the data– i.e., the balance between brain responses being driven by the stimulus (in a similar way across people) versus unrelated noise. One interpretation of our main findings (e.g., Fig. 2) is that this balance seems to change across the different cognitive circumstances reflected in the different experimental conditions in our study.

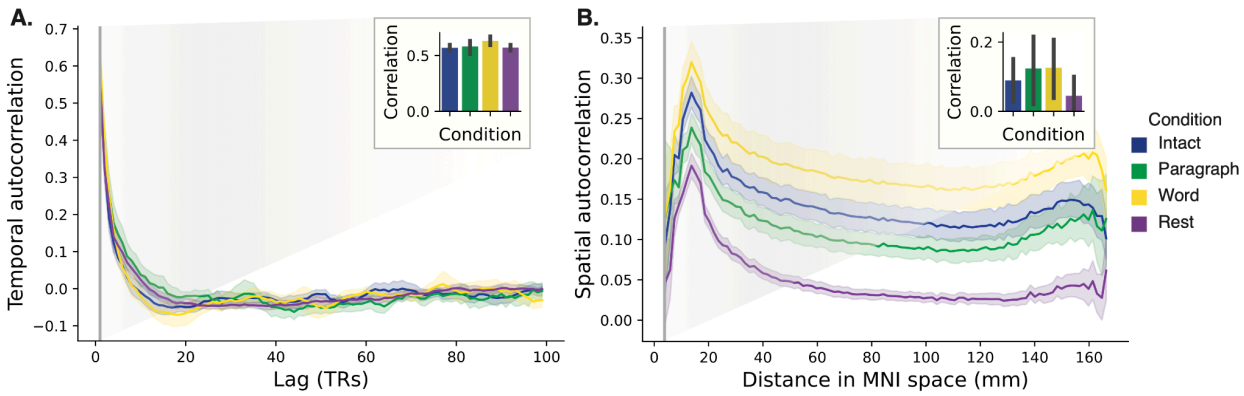
It is also important to note that observing or assuming that responses are time-locked to the stimulus does not necessarily mean that the responses are time-locked in a similar way across people. For example, in our prior work (e.g., Chang et al., 2021, Science Advances), we show that emotional responses (e.g., to very engaging stimuli) are often *not* time-locked across people, even though they are stimulus driven. In our current paper, we are specifically looking for responses that *are* similar across people, since our proxy for "informativeness" is across-participant decoding accuracy. In this way, our informativeness measure requires stimulus-driven time-locked responses (since we're essentially looking for responses that are similar across people), but stimulus-driven (time-locked) responses alone do not necessarily imply high "informativeness" in the way we are measuring it (i.e., the time-locked responses could differ across people).

Third, to our knowledge, the most similar study to ours is the Shinn et al. (2023, Nature Neuroscience) paper the reviewer points us to in their next comment. To our reading, the core point of that paper is that spatial and temporal autocorrelations reflect key network properties that change in meaningful ways across different cognitive or neurobiological circumstances. Our simulations (Fig. 1) vary spatial and temporal autocorrelations independently (in synthetic data), so in that sense the Shinn et al. (2023) paper has some overlap with ours. However, as we elaborate on in our response to the reviewer's next comment, although our measures of informativeness and compressibility can be *influenced* by temporal and spatial autocorrelations, respectively, those are not the only aspects of the data our measures reflect. A key advance over prior work is that the informativeness and compressibility measures in our study are specifically designed to get at representations of the *stimulus*, as opposed to solely being computable from properties of the neural activity (considering each participant independently) alone.

- In general, these simulations emphasize that spatial smoothing and temporal smoothing/autocorrelation should be major determinants of the results. This suggests two additional important considerations in my view: (1) would these simpler methods be a better way of discussing and characterizing these results (e.g., see Shinn et al, 2023)? And (2) to what extent will pre-processing factors influence the findings? For example, if a different basis set with different size kernels were used, how would the findings compare?

In the simulations, temporal smoothness is a "stand in" for how unique each time point's activity is, *in a way that is stimulus driven*. In other words, in the simulations, each "participant" is constructed by adding noise to a template pattern. More temporal correlations means that (depending on the amount of noise added) it becomes difficult to distinguish nearby timepoints.

In the fMRI data, the informativeness measure does *not* appear to be driven by temporal autocorrelations: autocorrelations are similar across the three listening conditions in the experiment, but the across-participant decoding accuracy (i.e., "informativeness") differs substantially. This can be seen most directly in the inset of Panel A in the Figure below (which we have added as Figure S1), which shows the lag 1 autocorrelations for each experimental condition:



Because the temporal autocorrelations are quite similar across conditions, the across-condition differences in our informativeness results cannot be driven by temporal autocorrelations. We explored this even further, by computing the autocorrelations at a range of lags (from 1 up to 100 TRs). As shown in Panel A of the figure, the temporal autocorrelations are also highly similar across every experimental condition within this range of lags.

There *do* seem to be some differences in spatial autocorrelations across the different experimental conditions, which could affect the compressibility results. To measure spatial autocorrelations, we computed (separately for each condition) the correlations between pairs of electrodes, as a function of their spatial distance in MNI space (measured in mm). When considering only the 1% of electrode pairs that were nearest to each other (inset in Panel B), there were no reliable differences in correlations across the different experimental conditions. However, as we correlate electrodes at farther distances, the specific changes in correlations as a function of distance *do* exhibit reliable differences across conditions (i.e., the curves for each condition in Panel B are separated as the inter-electrode distances increase). This tells us that spatial autocorrelations in the data change across the experimental conditions. However, the specific ways that spatial autocorrelations change across conditions (e.g., the rankings of different conditions' spatial autocorrelations) do not follow the ordering obtained using our compressibility measure, which tells us that (at the very least) compressibility is not measuring quite the same thing as spatial autocorrelation, even if the two measures might be partially related.

In summary, our new temporal and spatial autocorrelation analyses (Fig. S1) help to show that our informativeness and compressibility measures are not solely more complicated ways of characterizing temporal and spatial correlations in the data, respectively. In other words, although one *can* construct synthetic data (and presumably, find some real datasets) for which informativeness is solely related to temporal autocorrelations and compressibility is solely related to spatial autocorrelations, those simpler autocorrelation measures to not adequately characterize informativeness and compressibility in our dataset, as we are using those terms.

The reviewer also asks a good question about the role of preprocessing. We have not done a deep dive into different possible preprocessing pipelines, as this would be incredibly computationally expensive. However: since our measures of informativeness and compressibility can be influenced by temporal and spatial correlations in the data, as well as by other factors that might affect the temporal or spatial properties of the data, different preprocessing choices could affect our results, just as they do in nearly all other studies. Our intuition (though we have not tested this directly) is that preprocessing decisions that affect the data from all conditions in similar ways (e.g., introducing the *same* spatial or temporal blurring of the data from every condition) would lead to similar *qualitative* results with respect to the ordering of informativeness and compressibility across conditions. For example, blurring a time-varying signal might reduce its decoding accuracy (informativeness), but it seems unlikely that we would change the relative orderings (across *conditions*) of which signals were more vs. less informative via temporal blurring. Similarly, spatial blurring would likely affect the numerical (absolute) compressibility values we observed, but we don't see an obvious way that blurring out an already highly compressible signal would systematically lead to a *less* compressible signal– so again, the relative orderings across conditions should (we think) be preserved regardless of the precise kernel sizes. The above notwithstanding, since we did not test these intuitions, we have added a note (p. 18) acknowledging the possibility that our results may be affected by preprocessing choices, as in many neuroimaging studies.

- The authors' response on Fig. 3 changes over time seem not super well borne out by the results (while I agree they make intuitive sense). If you were to order quarters of the dataset by their peak decoding, intact would show Q3 > Q1 = Q4 > Q2. Paragraph shows Q2 > Q3 > Q1 = Q4. Word shows Q1 > Q4 > Q2 > Q3. So for 3/4 I don't really see the ordering effects that are being pointed to here in the text and the response. Rest is the only one to show a pattern with relatively consistent ordering, which I could see as being associated with an arousal effect on temporal auto-correlation over the course of the session, mixed in with a start of scan effect. Thus, unless better evidence is provided, I think this discussion should be removed. I also think the addition of the text to pg. 10 is good, but should more clearly outline the limitation (that the stimuli were always presented in the same order) and the confound (that therefore these results could be associated with stimulus specific effects).

The reviewer is correct that the peak decoding accuracies in the "quartile" analysis (Fig. 3A) are "out of order." We are basing our claims about the analysis on Figure 3C, which shows the changes in decoding accuracy over time (i.e., slope of the regression lines) as a function of the number of components, for each condition. The "intact" and "paragraph" conditions show positive slopes (i.e., decoding accuracy improves over time, on average, across a wide range of numbers of components– blue and green curves in Fig. 3C) whereas the "word" and "rest" conditions show negative slopes (i.e., decoding accuracies get *worse* over time, on average, across a wide range of numbers of components– yellow and purple curves in Fig. 3C).

We also take the reviewer's point that these findings may reflect something specific to *these* stimuli, as opposed to a more general phenomenon. We have added a note to this effect on page 10 of our revised manuscript.

- To clarify for the new permutation analyses associated with the Yeo parcellation: were these shuffled once per permutation, with orders kept consistent across participants? I'm surprised decoding accuracy would be so poor if so, given that there likely is information distributed across many brain regions. Shuffling separately per participant will break the relationship of brain regions across participants, and so seems like an inappropriate null for the current question (I.e., whether number of regions alone can explain the observed patterns)

To clarify, our goal with the permutation analysis was to test for the possibility that our observed correlations between decoding accuracy and network order reflected the numbers of nodes in each network, as opposed to something specific to the networks themselves. The permutation procedure we designed (p. 12, 23) used the same shuffled labels for all participants (we have added a note to this effect on page 23). The intuition is that the shuffled data retains the same numbers of nodes in each network, and the same "match" in labels across participants, but the specific locations of the nodes included in each "network" in the shuffled data were varied for each iteration of the procedure.

The "reduced" decoding accuracy in the shuffled data (for some of the shuffled brain networks, relative to the unshuffled networks) does not indicate that shuffling the network labels entirely breaks the ability to decode accurately. Rather, what we see when we shuffle the network labels is that the shuffled distributions approach something closer to the average across networks (e.g., the vertical midpoint of the scatterplots in Fig. 4D). We note that the shuffled accuracies are not *numerically* equal to the midpoints, since we use random assignments.

For the intact and paragraph conditions, since unshuffled high-order networks (e.g., default mode; pink dots in Fig. 4D) tend to be above this shuffled "average," whereas lower-order networks (e.g., limbic; dark blue dots in Fig. 4D) tend to be below this shuffled "average," the correlations between average decoding accuracy and network order is reliably higher in the intact versus shuffled data for these conditions (p. 12). We found the opposite pattern for the word and rest conditions, whereby we found stronger *negative* correlations between decoding accuracy and network order in the unshuffled vs. shuffled data (also reported on p. 12).

- I still think the abstract doesn't do a great job of setting up the research approach. The first two sentences are broader - but maybe too broad to clearly indicate what the research question is and why this is the right dataset to test it with. Why is depth of processing relevant to this question? I also think the significance statement is misleading, in its focus on ongoing cognition, since this paper explicitly focuses on stimulus-driven activity that is common and time-locked across people.

We have updated the abstract and significance statement to address the reviewer's suggestions.

Minor critiques:
- The glass brain display in Figure 4A is a poor visualization of brain networks. From this display, it seems as though the DMN is extremely extensive, relative to the published networks.

We have updated Figure 4 (panels A and B) to use a new rendering engine along with additional brain views that more accurately display each network (including the DMN).

- I still find Fig. 5 VERY hard to read. What is the primary take away that a reader should get from this figure?

There are several points that Figure 5 is intended to illustrate. Our main point is that the first principle components of the brain activity patterns were highly similar across all four experimental conditions (leftmost columns of panels A and C). This is true both with respect to the patterns themselves (panel C) and also the sets of functions that the components weigh on (per our reverse correlation analysis via Neurosynth; panel A). Other components seem to vary more substantially across conditions (right columns of panels A and C). We highlight these take-aways on page 12 when we introduce Figure 5.

To provide some additional context, a major reason that this figure is so complicated is that it is highly information dense. However, there are many potentially interesting observations to unpack, which (in our view) justifies the format of the figure. To help give a sense of how we think about that figure, here are some examples of "additional insights" that can be gained from the figure:

- **Consider the first principle component of activity across the different experimental conditions. In Panel A (leftmost sub-panel) we can see that the three "listening" conditions (blue, green, and yellow curves) are all more similar than the "rest" condition (purple curve). We can also use the color coded dots to see which *specific* functions are different across the listening vs. rest conditions. In general, the listening conditions weigh more heavily on factors related to cognitive control and language, whereas the rest condition weighs more heavily on factors related to spatial cognition, attention, motor control, and resting state.**
- **For any given function (by broad category summarized in the legend on the right, or by specific function described in the legend on the bottom) and component, we can use the radar plots to see how each condition weighs on that function. One interesting way to read the plots is to examine the relative weights given to a particular function across the different conditions. The second principal components seem to highlight a difference in language factors (magenta) and sensory perception factors (yellow-green) for the word-scrambled versus other conditions. The fourth principle component seems to group the paragraph and word conditions, and the intact and rest conditions. And so on.**

Zooming out, we see the consideration of how different components of neural activity weigh on different cognitive factors across different cognitive circumstances as a potentially promising future research direction. Figure 5 serves as a sort of "placeholder" or "jumping off point" for that future work. In the meantime, within the context of the current paper (where this analysis is not the primary focus), we have provided all of the data and code used to carry out the analyses we report, which could also give other researchers an opportunity to examine or test these questions further.

References

Shinn, Maxwell, et al. "Functional brain networks reflect spatial and temporal autocorrelation." Nature Neuroscience (2023): 1-12.

Reviewer #3:
Suitable Quality?:Yes
Sufficient General Interest?:Yes
Conclusions Justified?:Yes
Clearly Written?:Yes
Procedures Described?:Yes
Supplemental Material Warranted?:Yes

Comments on Significance Statement:
The significance statement is largely okay, but I think that defining compressibility as being robust to dimensionality reduction has too much jargon. Rewording this definition to appeal more to general audience would strengthen the statement.

We have re-worked the significance statement as suggested.

Comments:
The revised manuscript and response from the authors answers all the questions and comments from my original review. Notably, the newly added simulations clearly demonstrate the independent nature of compressibility and informativeness. And, the authors' detailed reply to the question of novelty relative to related prior work describes well the valuable contribution of the current work. So too, the additional discussion points of the current work's limitations help situate the paper in the literature and motivate important avenues for future related research. Overall, my initial excitement for this work has only been strengthened by the revised manuscript.

We thank the reviewer for their positive assessment of our work.